Midterm 2 will cover topics from 3.1, 3.2, 3.3, 3.4, 3.5, and 3.7.

- Problem 1 - Set up the appropriate form of partial fraction decomposition for a rational function. (Section 3.4)
- Problem 2 - Complete the arguments of using comparison test to determine if an improper integral is convergent or divergent. (Section 3.7)
- Problem 3 - Integrals using u-substitution, integration by parts, partial fractions or trigonometric substitutions. (Section 3.1, 3.2, 3.3, 3.4, 3.5)
- Problem 4 - Evaluate improper integral using limit notation. (Section 3.7)

1. Set up the partial fraction decomposition of the following rational function. DO NOT determine the numerical values of the coefficients. DO NOT integrate the function.

1). $\dfrac{2 - x + x^2 + 4x^3}{(x+4)^3(x^2+4)} =$

2). $\dfrac{x^2 - 2x + 5}{(x^2+9)x^2(x-1)} =$

3). $\dfrac{x^3 + 2x - 1}{(x^2+1)^2(x+2)} =$

2. For each of the following improper integral, one may use Comparison Test to determine if they are convergent or divergent. Circle the correct answers to complete the arguments of using the comparison test.

1). The improper integral $\displaystyle\int_1^\infty \frac{1}{\sqrt[7]{x^4} - 0.3}\,dx$ is (fill in "convergent" or "divergent") because $\dfrac{1}{\sqrt[7]{x^4} - 0.3}$ (choose one between A and B, and then choose one among C, D, E, and F).

A). $\geq$ B). $\leq$, C). $\dfrac{1}{x^4}$, D). $\dfrac{1}{x^7}$, E). $\dfrac{1}{x^{7/4}}$, F). $\dfrac{1}{x^{4/7}}$

2). The improper integral $\displaystyle\int_1^\infty \frac{x^{3/2}}{\sqrt{x^6 + 3}}\,dx$ is (fill in "convergent" or "divergent") because $\dfrac{x^{3/2}}{\sqrt{x^6+3}}$ (choose one between A and B, and then choose one among C, D, E, and F).

A). $\geq$ B). $\leq$, C). $\dfrac{1}{x}$, D). $\dfrac{1}{x^{3/2}}$, E). $\dfrac{1}{x^{1/6}}$, F). $\dfrac{1}{x^6}$

3). The improper integral $\int_1^\infty \frac{\sin^2 x}{\sqrt[3]{x^5+3}}\,dx$ is ____________ (fill in "convergent" or "divergent") because $\frac{\sin^2 x}{\sqrt[3]{x^5+3}}$ ______ ______ (choose one between A and B, and then choose one among C, D, E, and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x^3}$, D). $\frac{1}{x^5}$, E). $\frac{1}{x^{3/5}}$, F). $\frac{1}{x^{5/3}}$

3. Evaluate the following integrals.

1). $\int_0^{\pi/3} \sin^2\theta\cos^3\theta\,d\theta$

2) $\int \tan^3\theta\sec^4\theta\,d\theta$

3). $\int xe^{-2x}\,dx$

4). $\int \frac{\ln x}{\sqrt{x}}\,dx$

5). $\int \sin^3 x\cos^3 x\,dx$

6). $\int e^x\cos x\,dx$

7). $\int \arcsin(x)\,dx$

8) $\int \arctan(x)\,dx$

9). $\int x^2\ln x\,dx$

10). $\int \sin^2 x\tan x\,dx$

11). $\int \frac{\sqrt{x^2-4}}{x^2}\,dx$

12). $\int \frac{x^2}{\sqrt{9-x^2}}\,dx$

13). $\int \frac{x^2}{(1+x^2)^{3/2}}\,dx$

14). $\int \frac{1}{x^2\sqrt{x^2+16}}\,dx.$

15). $\int \frac{1}{x^3\sqrt{x^2-1}}\,dx$

16). $\int_0^1 \frac{x^2}{(4-x^2)^{3/2}}\,dx$

17). $\int \frac{3x+6}{x^2-3x}\,dx$

18). $\int \frac{4x+2}{x^2-2x-8}\,dx$

19). $\int \frac{4x^2-x+8}{x(x^2+4)}\,dx$

20). $\int \frac{5x^2+8}{(2x+1)(x^2+9)}\,dx$

4. Evaluate the following improper integrals or show that they are divergent.

1). $\int_1^\infty \frac{(\tan^{-1}x)^2}{x^2+1}\,dx$

2). $\int_1^4 \frac{1}{(x-1)^{2/3}}\,dx$

3). $\int_1^\infty \frac{3}{(x+1)^3}\,dx$

4). $\int_{-2}^0 \frac{1}{(x+2)^2}\,dx$

Formula Sheet

$$\sin^2 x + \cos^2 x = 1, \quad \sin^2 x = 1 - \cos^2 x, \quad \cos^2 x = 1 - \sin^2 x$$

$$\sec^2 x = \tan^2 x + 1, \quad \tan^2 x = \sec^2 x - 1$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}, \quad \sin 2x = 2 \sin x \cos x$$

$$\sin A \cos B = \frac{1}{2}[\sin(A - B) + \sin(A + B)]$$

$$\cos A \cos B = \frac{1}{2}[\cos(A + B) + \cos(A - B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\int \cot x \csc x \, dx = -\csc x + C, \qquad \int \csc^2 x \, dx = -\cot x + C$$

$$\int \tan x \, dx = \ln|\sec x| + C, \qquad \int \cot x \, dx = \ln|\sin x| + C$$

$$\int \sec x \, dx = \ln|\sec x + \tan x| + C, \qquad \int \csc x \, dx = \ln|\csc x - \cot x| + C$$

$$\int \frac{1}{a^2 + x^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C, \qquad \int \frac{1}{\sqrt{a^2 - x^2}} = \arcsin\left(\frac{x}{a}\right) + C$$

$$\int_1^\infty \frac{1}{x^p} \, dx = \begin{cases} \text{is convergent} & \text{if} \quad p > 1 \\ \text{is divergent} & \text{if} \quad 0 < p \le 1. \end{cases} \qquad \int_0^1 \frac{1}{x^p} \, dx = \begin{cases} \text{is convergent} & \text{if} \quad 0 < p < 1 \\ \text{is divergent} & \text{if} \quad p \ge 1. \end{cases}$$